A Report on

Turbulent Diffusion

Student Name: *ID Number:*

Vaishak P 2012B5PS399H

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Name: Vaishak P *Name of guide:* Prof. K.Subramanian
Discipline: Physics *Name of PS Faculty:* Dr. S.K.Sahay

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ABSTRACT

The project will involve first reading the basics of fluid dynamics, then looking at the concept of turbulence in fluids. After basic reading, turbulent diffusion of a passive scalar field, like the concentration of smoke, will be studied. If enough progress is made,the student could also write a code to follow the trajectories in a simulation of turbulence, to simulate turbulent diffusion.

Signature of the Student: *Signature of the PS Faculty:*

Date:

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1 Introduction

Fluid dynamics is the study of flows of fluids. A fluid is mathematically a continuum,with no constituents i.e it is not seen to be composed of discrete molecules.Hence calculus can be used. But because of the tiny size of molecules and very large numbers of them, we can approximate real fluids as a continuum on a suitable length-scale (the lower limit of the infinitesimal spatial quantities used in limits i.e in spatial derivatives).

2 Properties of flows

Some properties of flow which describe its state are:

- Density
- Velocity of an infinitesimal element
- Forces between elements(pressure, shears, etc)

However, there might be more properties to add like temperature, heat flow (If we wish to study conduction), currents and electromagnetic fields, which have not been studied in this project.

2.1 Continuum approximation

Mass density is defined in a particular range of length scale, in which it is almost a constant, below which it no longer varies smoothly with the length scale due to molecular discreteness which is to be avoided to achieve a continuum approximation.

Velocity of an element is the average velocity of its constituents (i.e velocity of Center of mass of the element).The length scale is such that the velocity of the constituents do not vary very much throughout the cube.

To describe pressure, we first make the assumption that the fluid element is in *local thermodynamic equilibrium*. i.e it has, approximately, an isotropic pressure and is 'almost' in equilibrium locally i.e pressure varies infinitesimally over the chosen length scale.

So we can describe the density ($\rho(\vec{x},t)$), pressure ($p(\vec{x},t)$) as a continuous scalar variables and velocity ($\vec{v}(\vec{x},t)$) as a vector field.

2.1.1 State of local thermodynamic equilibrium

In a fluid element, the velocity of its constituents can be decomposed into a center of mass part and a random ($\vec{v}_r = \vec{v} - \vec{v}_{com}$) component. In the COM frame of the element, the fluid constituents only have the random part.Even though neighboring elements are in motion relative to the COM of the central element, we assume that all the neighboring elements are in almost thermodynamic equilibrium, manifesting as isotropic pressure, 'almost' because the equilibrium state varies from point to point infinitesimally, permitting pressure to be a continuous variable.

In any gas, $(v_{rx})^2 = (v_{ry})^2 = (v_{rz})^2 = \frac{\vec{v_r}^2}{3}$ because different degrees of freedom interact through collisions. This is because collisions can be oblique, kinetic energy gained due to fall in any potentials present is distributed (equally) in all directions of KE and interacting degrees of freedom. This manifests as isotropic pressure.

3 Laws of Fluid Mechanics

Any element of a fluid is just a chosen boundary that defines it. It is just a point of view we are considering for the sake of interpretation (like any infinitesimal volume element has different shapes in different coordinate systems,all unifying in the limit). The fluid behavior and its equations must be independent of the coordinate system chosen to describe it. Hence the equations must be vector equations.

Consider one component of force due to pressure on a volume element $d\tau = dA_x dx$, defined by its boundaries.It contains a mass $\rho d\tau$.The same mass after a time dt is transported a distance $\vec{v}(\vec{x},t)dt$ and is contained in a volume element $d\tau'$ of different shape and size. The components of forces which seem to cancel off (the smaller of p(x) and p(x+dx)) are responsible for the changing shape and size (if compressible) of the volume element. But the mass contained doesn't change.Hence,

$$\rho(\vec{x},t)d\tau = \rho(\vec{x}+\vec{dx},t+dt)d\tau' \tag{1}$$

$\vec{dx}$ is related to dt through $\vec{dx} = \vec{v}dt$. Hence, we are trying to calculate the full time derivative of a field quantity as seen by an observer. moving with the particle i.e moving with velocity $\vec{v}$ at $\vec{x}$.Let the quantity be $Q(\vec{x},t)$.Then,

$$\begin{aligned} Q(\vec{x}+\vec{dx},t+dt) - Q(\vec{x},t) &= \frac{dQ}{dt}dt \\ &= \frac{\partial Q}{\partial t}dt + \vec{dx}\cdot\nabla Q(\vec{x},t) \\ &= \frac{\partial Q}{\partial t}dt + \vec{v}\cdot\nabla Q(\vec{x},t)dt \end{aligned}$$

The full time derivative of Q is called the Lagrangian derivative:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \vec{v}\cdot\nabla$$

Returning to mass conservation, by chain rule or Taylor expansion of RHS of (1),

$$\begin{aligned} \rho(\vec{x}+\vec{dx},t+dt)d\tau' - \rho(\vec{x},t)d\tau = \frac{d}{dt}(\rho d\tau)\,dt &= 0 \\ \Rightarrow \frac{d}{dt}(\rho d\tau) &= 0 \qquad (2) \\ \Rightarrow \frac{d\rho}{dt}d\tau + \rho\frac{d}{dt}(d\tau) &= 0 \qquad (3) \end{aligned}$$

We now need to find the second term in (3).Consider a cube of volume $d\tau$ as shown.

Different faces are moving with different velocities thereby changing volume.The rate at which volume is changing is

$$v^i(x_i+dx_i)dA_i - v^i(x_i)dA_i = \nabla\cdot\vec{v}d\tau$$

Hence,

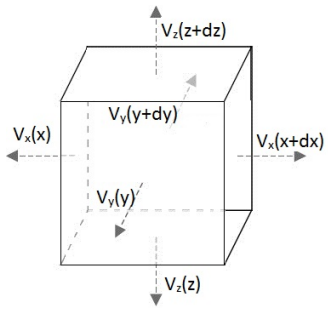


Figure 1: *An expanding cube*

$$\frac{d}{dt}(d\tau) = \nabla \cdot \vec{v} d\tau$$

Inserting in (3) we get

$$\nabla \cdot (\rho \vec{v}) + \frac{\partial \rho}{\partial t} = 0$$

Which is the conservation equation for mass.

Since $p(x)$ $p(x+dx)$, the pressure at any point corresponds to the present compression state of the volume element, if it is compressible. If it is incompressible, then it behaves as a rigid body in the sense of effectively communicating any force acting on other points any distance away from the considered element instantaneously(i.e we have chosen speed of sound waves to be infinity).

3.1 Newton's Law

The force equation for this element is

$$\frac{d}{dt}(\rho \vec{v} d\tau) = \vec{F}$$

Or, from (2)

$$\rho d\tau \frac{d\vec{v}}{dt} = \vec{F}$$

F_x due to pressure is

$$p(x)dA_x - p(x+dx)dA_x = -(\nabla p)_x d\tau$$

Also adding force due to gravity (self gravity or an external field),

$$\vec{F} = -\nabla p d\tau - \rho \nabla \phi d\tau + otherf d\tau$$

Where 'other f' is other force per unit volume that may be present. Hence the equation of motion is

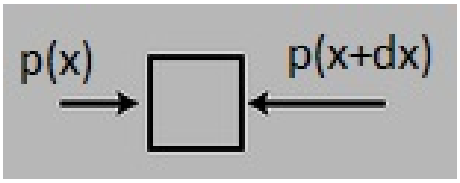


Figure 2: *Force due to pressure on a cube*

$$\rho \frac{d\vec{v}}{dt} = -\nabla p - \rho \nabla \phi + f \tag{4}$$

Which is the Euler's equation.

3.2 Newton' Law in Conservation Form

Consider a fixed infinitesimal cube in space through which fluid flows in and out. When there are no external forces (i.e when RHS of (4) is 0), Momentum of fluid must be conserved. Momentum in the fixed cube is changing because the fluid which carrying momentum is flowing in and out. The net momentum change due to fluid entering and leaving in the i direction is (like $\frac{\partial}{\partial t}(m)\vec{v}$):

$$(\rho v_i \vec{v} dA_i)(x_i + dx_i) - (\rho v_i \vec{v} dA_i)(x_i) = (\nabla \cdot T)_i d\tau$$
$$= \text{rate of decrease of momentum inside} = -\frac{\partial}{\partial t}(\rho \vec{v})\, d\tau$$

Where $T_{ij} = \rho v_i v_j$.Hence,

$$\nabla \cdot T + \frac{\partial}{\partial t}(\rho \vec{v}) = 0$$

If there are external forces (per unit volume) present, then the equation will have an inhomogeneous(source) term on the RHS instead of 0, making it in our case the Euler equation.

$$\nabla \cdot T + \frac{\partial}{\partial t}(\rho \vec{v}) = -\nabla p - \rho \nabla \phi$$

3.2.1 Bernoulli's Theorem

Consider a fluid flow which is incompressible, steady and has constant density along all streamlines. Properties of the fluid element at 1 and 2 are ($p_1, \rho_1, \vec{v}_1, dA_1$ and $p_2, \rho_2, \vec{v}_2, dA_2$) .Define that part of the fluid shown as the system.Let us name different elements of the fluid in the system from 1 to n. Since the flow is steady, 1

replaces and takes the state of 2, 2 that of 3...n that of n+1. The net change in the system is that element 1 disappears and element n+1 appears. There is a change in kinetic energy of the system which is KE(n+1)-KE(1). The work done in time dt on 1 by fluid exterior to the system and in contact with it is. $p(1)dA_1\vec{v}_1dt$ Similarly on n is $p_(n+1)dA_n\vec{v}_ndt$.

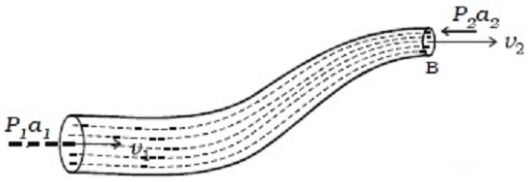


Figure 3: *A stream tube*

Hence,

$$p(1)dA_1v_1dt-(p(n+1)dA_nv_n)dt=(\frac{1}{2}\rho_2\vec{v}_2^2dA_2v_2dt-\frac{1}{2}\rho_1\vec{v}_1^2dA_1v_1dt)+(\rho_2\phi_2dA_2v_2dt-\rho_1\phi_1dA_1v_1dt)$$

Also since $\rho_1dA_1v_1=\rho_2dA_2v_2$,

$$\frac{1}{2}\vec{v}_1^2+\frac{p_1}{\rho_1}+\phi_1=\frac{1}{2}\vec{v}_2^2+\frac{p_2}{\rho_2}+\phi_2$$

The apparent change in the system is that 1 disappears and n appears. Hence if we are ignorant of the flow of elements between 1 to n, it would (faultily) look as though causality is violated i.e 1 is converted to n in time dt, independent of the distance from 1 to n.

That is,if there are symmetries or wrt certain observable variables if the state of the 'fluid' elements from 1 to n are indistinguishable, then causality is apparently violated in the observable phenomena described by these observable variables.

More general version

$$\rho\frac{d\vec{v}}{dt}=-\nabla p-\rho\nabla\phi$$

For steady flows, $\frac{\partial}{\partial t}(\vec{v})=0.\rho\frac{d}{dt}(\vec{v})$ gives force. So $\rho\frac{d}{dt}(\vec{v}d\tau)\cdot\vec{v}$ gives rate of work along pathline (streamline).

Hence,

$$\rho\frac{d\vec{v}}{dt}\cdot\vec{v}=-\vec{v}\cdot\nabla p-\rho\vec{v}\cdot\nabla\phi$$
$$\Rightarrow\rho\vec{v}\cdot\nabla(\frac{1}{2}\vec{v}^2+\phi)=-\vec{v}\cdot\nabla p \quad (5)$$

Consider

$$\rho\vec{v}\cdot\nabla\frac{p}{\rho} = -\frac{p}{\rho}\vec{v}\cdot\nabla\rho + \vec{v}\cdot\nabla\rho$$

also,from continuity equation

$$\begin{aligned}\nabla\cdot(\rho\vec{v}) &= 0\\ \Rightarrow \vec{v}\cdot\nabla\rho &= -\rho\nabla\cdot\vec{v}\end{aligned}$$

Hence (5) becomes

$$\rho\vec{v}\cdot\nabla(\frac{1}{2}\vec{v}^2 + \phi + \frac{p}{\rho}) = \frac{p}{\rho}\nabla\cdot\vec{v}$$

$$\begin{aligned}\text{If} \qquad PV^{\chi} &= c,\\ \nabla\cdot\vec{v} &= -\frac{1}{V}\frac{d}{dt}(V)\\ &= \theta\\ &= -\frac{1}{\rho\chi}\vec{v}\cdot\nabla p\end{aligned}$$

The RHS terms of the previous equations arise only due to compression of the fluid element i.e. when ρ is not a constant along the streamline in a steady flow.
If

$$\begin{aligned}\theta &= 0,\\ \vec{v}\cdot\nabla\rho &= 0\end{aligned}$$

Or, each streamline behaves as a tube (with varying cross section) with rho varying only with tube.

4 Vorticity

Vorticity is the phenomenon of whirlpools i.e some sort of circulation around a point. Its can be defined precisely.

we wish to calculate the angular velocity with which a weightless stick of infinitesimal length would rotate, when placed at the chosen point.

A volume element cannot rotate about itself unlike solids, since it cannot resist the centrifugal forces by internal forces. A difference in the parallel component of velocity between two opposite faces just means that the fluid at the faces are sliding past each other.

Consider a viewpoint as shown. Let v_{avg} be the average tangential velocity of all elements around a circle of radius r centered at the chosen point.i.e

$$\begin{aligned} v_{avg}(\vec{x},r,t) =& \frac{\oint \vec{v}\cdot d\vec{l}}{\pi r^2} \\ =& \lim_{r\to 0} \frac{2v_{avg}}{r} \\ =& \nabla\times\vec{v} = \omega \end{aligned}$$

Where $\frac{\omega}{2}$ is the angular velocity of the rod.

4.1 Angular Momentum

The pressure in an inviscid fluid is isotropic at any point.This means that the net force on any imaginary surface is always normal to it.Hence the angular momentum of any volume element is conserved since there are no tangential forces (torques). Consider a cylinder of fluid of cross sectional area dA_1 and length dl_1.

$$\delta\frac{\Delta\vec{L}}{\Delta\tau} = \rho\frac{\vec{\omega}}{\pi}\cdot dA$$

Hence, $\int \vec{\omega}\cdot dA$ is a measure of $\frac{\Delta\vec{L}}{\Delta\tau}$ of a sheet of fluid S when ρ is constant.Due to conservation of $\vec{L}$,

$$\vec{\omega}_1\cdot dA_1(t) = \vec{\omega}_2\cdot dA_2(t+dt)$$

as the vortex gets transported from 1 to 2.In integral form over some surface S with its boundary B,

$$\frac{d}{dt}\left(\int_S \vec{\omega}\cdot dA\right) = 0$$

Rate of change in integral due to change in $\vec{\omega}$ with same area + change due to rate of change of area at the boundary with same $\vec{\omega}$

$$
\begin{aligned}
&= \int_S \frac{\partial \vec{\omega}}{\partial t} \cdot dA + \oint_B (\vec{v} \times d\vec{l}) \cdot \vec{\omega} \\
&= \int_S \frac{\partial \vec{\omega}}{\partial t} \cdot dA + \oint_B (\vec{\omega} \times \vec{v}) \cdot d\vec{l} \\
&= \int_S (\frac{\partial \vec{\omega}}{\partial t} + \nabla \times (\vec{\omega} \times \vec{v})) \cdot dA = 0
\end{aligned}
$$

Identically.

$$
\Rightarrow \frac{\partial \vec{\omega}}{\partial t} + \nabla \times (\vec{\omega} \times \vec{v}) = 0
$$

This along with

$$
\begin{aligned}
\nabla \times \vec{v} &= \vec{\omega} \\
\nabla \cdot \vec{v} &= 0
\end{aligned}
$$

completely determine the velocity field and hence the flow in the inviscid case.

5 Viscous Flow

As mentioned earlier, there exist relative velocities of neighboring elements w.r.t any considered element. This causes collisions at the surfaces of interface, the momentum transfer occuring in which causes viscous forces. This friction not only changes bulk velocity of the participating elements but some portion of the bulk kinetic energy possessed goes into random thermal energy.

Vaguely, the mechanism might be as follows. Force on 1 due to 2 is due to collision of constituents (molecules) in 1 and 2 at the surface.The collisions which produce the viscous force is due to the relative excess bulk momentum of 2.(The remaining random part does not come into picture due to the assumed local thermodynamic equilibrium) Hence i direction of force between 1 and 2 is proportional to $\frac{\partial}{\partial x_j}\left(v_i(x_j)\right)$ The force on 3 due to 2 is similar equation but evaluated at $x_j + dx_j$.(summation over j is intended).Since the force on 2 due to 3 is negative of this, the net force on 2 is the difference of these two which is proportional to second derivative of v_i i.e

$$F_i \text{ is proportional to } \quad \nabla^2 v_i$$

5.1 Stress tensor

The stress tensor can be defined at each point and is interpretable as the force per unit area on the 3 faces containing the vertex at which this is defined. For The

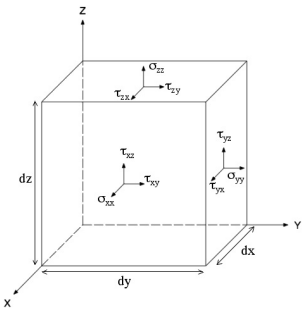


Figure 4: *Viscous forces on a cube*

definition of the viscous tensor please refer to [1]

And also for the derivation of the stress strain relation see [3] Where it is given in detail. Since the cube is not rotating about itself, the net torque is 0. One can show that this implies

$$S = S^T$$

where S is the viscous stress tensor. The viscous force is

$$f_s = -\nabla \cdot S$$

5.1.1 The Stress-Rate of Strain relation

Decomposition of the velocity gradient

The velocity gradient $V = \nabla\vec{v}$ can be decomposed into a pure divergence (θI), a shear (σ) and a curl (C).

$$\sigma = \frac{V + V^T}{2} - \frac{\theta}{3}$$

Where

$$\theta = \nabla \cdot \vec{v}$$
$$C = \frac{V - V^T}{2}$$

The viscous force is then related to the velocity gradient through

$$S = -2\eta\sigma - \alpha\theta I$$

where I is the identity tensor.

5.2 The Navier-Stokes equation

Adding the viscous contribution $-\nabla \cdot S$,the Euler equation with viscous force now becomes

$$\rho\frac{d\vec{v}}{dt} = -\nabla p - \rho\nabla\phi + \eta\nabla^2\vec{v} \tag{6}$$

This non linear equation is the dynamical equation for the flow of a viscous fluid. We usually are interested only in incompressible flows with constant density.

If we are interested only in the velocity field, then we can get rid of pressure and gravitational forces by taking curl of (6) to obtain:

$$\frac{\partial\vec{\omega}}{\partial t} + \nabla \times (\vec{\omega} \times \vec{v'}) = \nu\nabla'^2\vec{\omega} \tag{7}$$

We can solve for pressure by taking the divergence of (6). This gives:

$$\nabla \cdot \frac{\partial\vec{v}}{\partial t} + \nabla \cdot \vec{v} \cdot \nabla\vec{v} = \frac{-\nabla^2 p}{\rho} - \nabla^2\phi + \nabla \cdot \nu\nabla^2\vec{v}$$

The last term and the first are zero due to incompressibility. Hence,

$$\nabla^2 p = -\rho\nabla^2\phi - \rho\nabla \cdot (\vec{v} \cdot \nabla\vec{v})$$

This is an instance of Poisson's equation.

5.2.1 Dissipation due to viscosity

Now that we know the force due to viscosity, the rate of work by this force gives the rate of dissipation of mechanical energy per unit volume or mass.

Suppose we have a volume V of fluid whose boundary is A. A boundary area element is then

subjected to the viscous force $S \cdot dA$. Rate of work is

$$\begin{aligned} \dot{W} &= -\int_A \vec{v} \cdot S \cdot dA \\ &= -\int_A v_i S_{ij} dA_j \end{aligned}$$

By divergence theorem, this becomes

$$= -\int_A \frac{\partial}{\partial x_j} \left(v_i S_{ij} \right) d^3x$$

Hence, per unit volume, the rate of dissipation is

$$\begin{aligned} \dot{w} &= -\frac{\partial v_i}{\partial x_j} S_{ij} - v_i \frac{\partial S_{ij}}{\partial x_j} \\ &= -\frac{1}{2}(S_{ij}\frac{\partial v_i}{\partial x_j} + S_{ji}\frac{\partial v_i}{\partial x_j}) - v_i \frac{\partial S_{ij}}{\partial x_j} \\ &= -\frac{1}{2}(S_{ij}\frac{\partial v_i}{\partial x_j} + S_{ij}\frac{\partial v_j}{\partial x_i}) - v_i \frac{\partial S_{ij}}{\partial x_j} \\ &= -S_{ij}\sigma_{ij} - v_i \frac{\partial S_{ij}}{\partial x_j} \end{aligned}$$

Thus, the rate of work by the viscous forces consists of two terms. The second term is

$$-\vec{v} \cdot (\nabla \cdot S) = -\vec{v} \cdot (\nu \nabla^2 \vec{v})$$

which is the rate of increase of the mechanical energy of the fluid element. Thus the first term gives the rate of energy input into the random motions of the fluid element experiencing this force. Hence, the rate of dissipation per unit mass is,

$$\dot{w} = -S_{ij}\sigma_{ij} = 2\nu\sigma_{ij}\sigma_{ij} \tag{8}$$

5.3 Solutions to viscous flows

Suppose we want to solve the NS equation for given boundary conditions. The boundary conditions at least contain information about the geometry of the problem. One more fact which is experimentally seen to be true in low velocity flows (which are smooth and laminar) is that the velocity of the fluid is zero at the boundaries. Using these boundary conditions, man has been able to solve the NS equations for a few cases. Some examples wherein solution to the Navier-Stokes equation can be found in closed form are:

- Inviscid flow past an infinite cylinder
- Inviscid flow past a sphere
- Viscous flow past a sphere
- Laminar viscous flow inside a pipe (Poiseuille flow)

Some of these have direct analogues in electrostatics and hence the solution is borrowed from solutions for electromagnetic fields around dielectric spheres, etc. Solutions to these problems are given in [2] and [3].

Even these solutions are just 'well behaved' solutions of a given geometry. For the same geometry, we can have solutions which don't even look like the laminar ones theoretically obtained. So it seems that, for a particular geometry, there exist many possible flows. The fact is, there exists a continuous parameter set of solutions, explained below.

6 Turbulence

6.1 Some Flows

It would be useful to see how real flows, which must really be solutions of NS equations, behave. So, to have an overview of how complex phenomena a nonlinear equation like the NS equation can possess, it is worthwhile to look at some flows.

6.1.1 Flow past a cylinder

(This has been learnt from [8],[3] and also[5] and reproduced here. Refer to them for a detailed account)

Suppose we are analyzing flow past an infinite cylinder. Let its diameter be D, velocity far away from the sphere V_0 in the x direction. Let us use variables measured in characteristic quantities in the problem:

$$x' = \frac{x}{D}, y' = \frac{y}{D}, z' = \frac{z}{D}$$
$$\vec{v}' = \frac{\vec{v}}{V_0} \qquad t' = \frac{tD}{V}$$

If we now rewrite the vorticity equation (7) in terms of primed quantities, we get:

$$\Rightarrow \frac{\partial \vec{\omega}'}{\partial t'} + (\vec{\omega}' \times \vec{v}') = \frac{1}{\Re} \nabla'^2 \vec{\omega}'$$

where

$$\Re = \frac{\rho}{\eta} VD$$

is called the Reynolds number. It characterizes the flow. All solutions of are then of the form

$$\vec{v}' = \vec{v}'(\vec{x}', \Re)$$

This shows that for the same geometry, there exists a type of flow for every value of $\Re$. This is the continuous parameter of the set of solutions to a particular geometry.

This also permits us to build scaled down models of airplanes in such a way that the Reynold's number doesn't change and test them in the wind tunnel during its development to see how exactly the actual airplane behaves when built. Since $\Re$ is the same, the nature of the flow is exactly the same in the wind tunnel around the model.

Symmetries in the problem: Let us look at symmetries available in the problem. Any symmetry in the flow means that the NS equations and the boundary conditions are invariant under it. If this is true, then a flow with a particular symmetry is a possible solution and must exist in reality. Exactly when a particular type of flow may be realized will depend on the parameter $\mathfrak{R}$. Thus allowed symmetries may not be exhibited in a particular realization.

Possible symmetries present in this geometry are:

1. z invariance. The cylinder in infinitely long. i.e

$$\frac{\partial}{\partial z}(\text{any quantity}) = 0$$

2. Reflection in y.

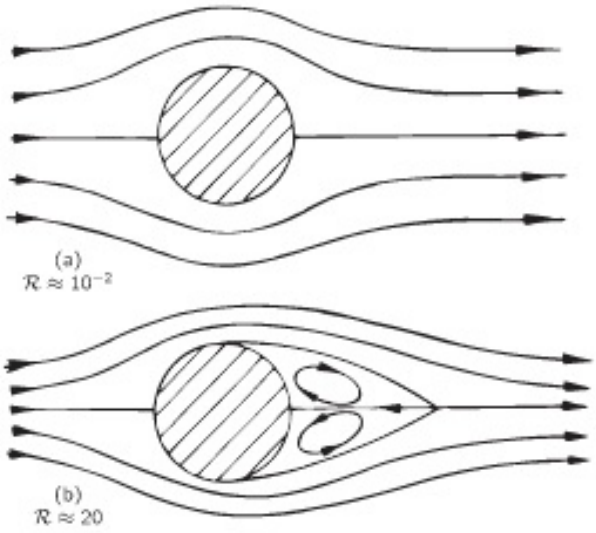
$$(x, y, z) \to (x, -y, z)$$
$$(v_x, v_y, v_z) \to (v_x, -v_y, v_z)$$

3. Time reversal, only when $\nu\nabla^2\vec{v} = 0$.

$$t \to -t$$
$$\vec{v} \to -\vec{v}$$

   Time reversal is the same as reflection in x, a composition of reflection in ($\vec{x}$ and $\vec{v}$) and reflection in (y) given above i.e reverse flow along the same pathlines as the original flow. The only difference being that sources now act as sinks and hence fluid flows from right to left.

Let us now look at realizations of flow for different values of $\mathfrak{R}$ and see at what $\mathfrak{R}$ what symmetries are exhibited



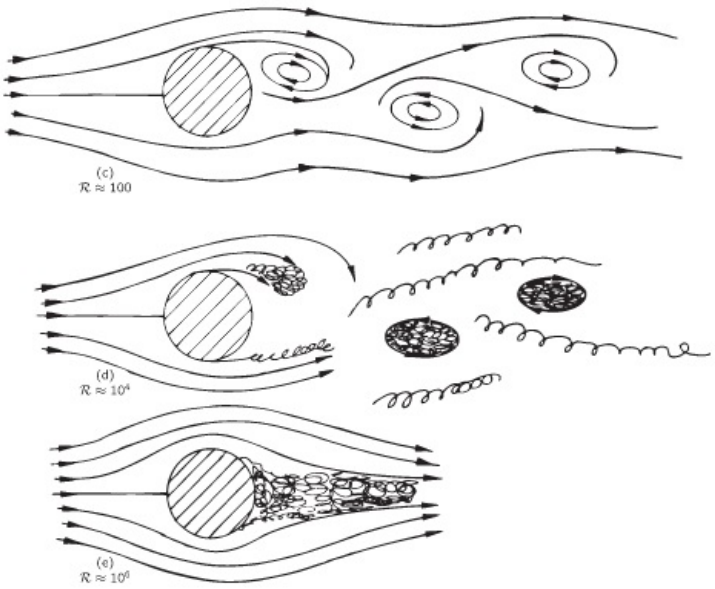


Figure 5: Transition to turbulence as $\mathfrak{R}$ is increased. Taken from [3]

In (a), $\mathfrak{R}$ is very small and hence is $\vec{v}$ and $\nu\nabla^2\vec{v} = 0$. So the flow is nearly x symmetric i.e nearly time reversal symmetric. This broken symmetry is easily noticeable in (b). (1) and (2) still exist, but (b) has departed far from (3).

In (c), (2) is broken and has a time period over which the whole pattern of flow repeats.However, (2) is now obeyed on an average over this period T_1. Some authors like [9] state that at around $\mathfrak{R} = 74$, the (1) is also broken. Also, (c) is different from (a) and (b) in the sense that (c) is unsteady flow. Velocity at any point varies with time, albeit periodically, with a finite period. As $\mathfrak{R}$ gets bigger, the flow becomes chaotic and smaller vortices appear and are shred. There is still a periodic motion of the (c) type superimposed on this turbulent one (see (d)). Then there comes a point when flow lines leave the cylinder i.e the boundary conditions change. The flow is chaotic, irregular and vortices of various sizes are created and shred.

6.1.2 Couette flow

(This has been learnt from [3] and reproduced here. Refer to it for a detailed account)

Couette flow is the flow between two concentric rotating cylinders. Oil is poured between the cylinders and fine aluminum powder is used for visualizing the flow. Let the outer cylinder remain stationary while the inner be made to rotate.

Successive features that emerge as the relative velocity of the two cylinders is increased are:

- Expected laminar flow with z invariance
- At a certain critical velocity, toroidal vortices occur, with azimuthal symmetry. Here, the continuous z symmetry is broken and is replaced by a discrete one

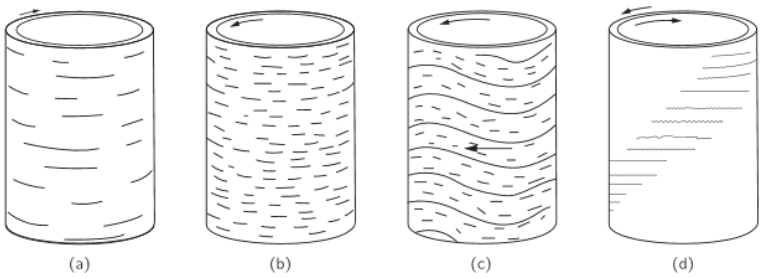


Figure 6: Couette flow, taken from [3]

due to these *Taylor cells* (see fig (7)). The net motion of a fluid particle is now helical, confined to a toroidal surface (b)

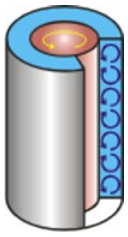


Figure 7: Taylor cells, the second phase of couette flow.

- The Taylor toroids become wavy, and continuous azimuthal symmetry is lost for an average one. Flow until now even though complex, is still non chaotic.

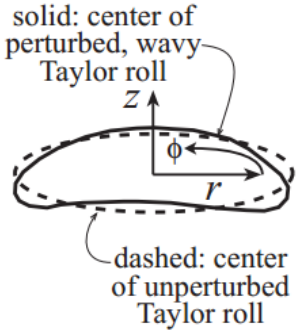


Figure 8: Wavy Taylor cells, the third phase of couette flow. Taken from [5]

- The flow then becomes irregular and turbulent (e).

What we see from these two flows, past a cylinder and couette, is that symmetries allowed by the geometry and NS equation are successively broken as $\Re$ is increased. Periodicities are present, a time average over which symmetries are preserved. Then many periods start to occur simultaneously until they are no longer commensurate with each other, and the symmetries are no longer preserved on an average over finite time periods. Thereafter, the flow becomes completely chaotic, i.e turbulent.

In fully developed turbulence, vortices of many sizes are present. Another observation is that fully developed turbulence looks the same irrespective of the geometry that produced it. Figure 9 shows turbulence behind a grid. This is not different from turbulence behind a cylinder or that between cylinders, if we do not consider the turbulence being carried by the mean flow.

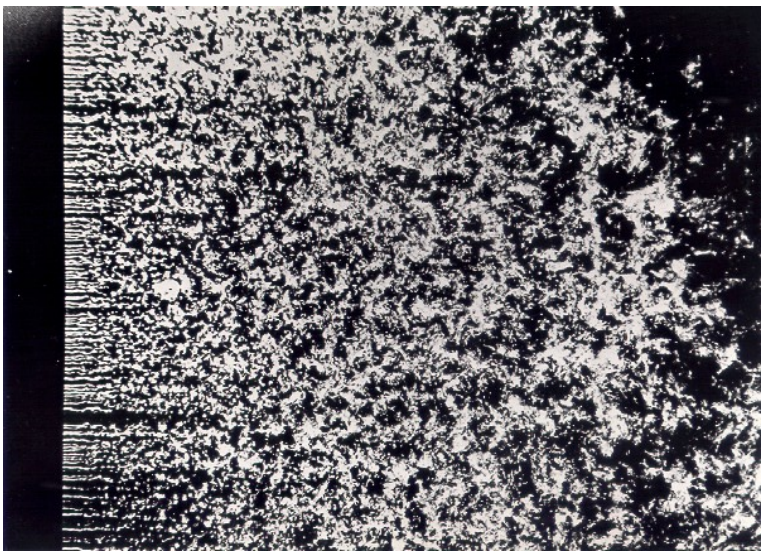


Figure 9: Turbulence behind a grid.

This might have motivated brilliant scientists to speculate that fully developed turbulence was a 'universal state'. On assumptions of statistical homogeneity and isotropy, a very rough theory (dimensional analysis) which is quite accurate in explaining some properties of turbulence.

In [10], a definition of turbulence is attempted. It says,

Incompressible hydrodynamic turbulence is a spatially complex distribution of vorticity which advects itself in a chaotic manner...The vorticity field is random in both space and time, exhibits a wide and continuous distribution of length and time scales.

6.2 Nature of a possible theory of turbulence

Having seen what turbulence is, it would now be nice if one could describe it. Even though we already have the NS equation which is supposed to completely describe any flow, it is mostly unsolvable. Furthermore, all complex flows have common properties:

1. the velocity field fluctuates randomly in time and is highly disordered in space.

2. the velocity field is chaotic in the sense that a minute change in the initial condition of a fluid element will produce a large change in its subsequent motion.

The randomness and sensitivity to initial conditions of the velocity field can be illustrated by following consideration. Suppose we measure a component of the

local velocity $\vec{v}(\vec{x}_0,t)$ and plot it with time in a realization of the flow (a particular run of a predefined flow). This is random and is shown in fig(10). In a second

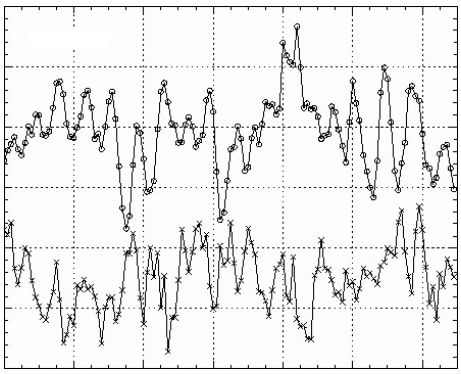


Figure 10: $v_i(\vec{x}_0,t)$ vs t for two different realizations.

realization, the same plot at the same point gives a different graph, even though it is the same experimental setup with the initial conditions (within experimental tolerance). Each realization would give a different graph, so $\vec{v}$ is not a reproducible quantity. This happens because, each time the experiment is carried out, the initial conditions are not exactly the same. They vary very slightly. The non linearities in the system then amplify these slightest variations to give the above effect.

But what are reproducible are the statistical properties of the flow. Some types of statistical averages which will be used are given below :

- **Ensemble average:** Suppose we were to carry out 10000 identical experiments and average $\vec{v}$ at each position and time over these realizations. This average is called the *ensemble average* and is denoted by $\langle u \rangle$.

- **Time averages and Space averages:** usual time average denoted by $\overline{u}$ over some chosen time scale.

- **Space averages:** usual space average over some chosen length scale i.e $\frac{1}{L^3}\int u d^3x$

The statistical averages of quantities are seen to be reproducible. For instance, consider $\langle v_i^2 \rangle$ (fig(11), the ensemble average of a squared velocity component at a particular point in space and at an instant of time. It is a smooth function of space and time unlike v_i^2 itself. So is $\overline{v_i^2}$ and its spatial average.

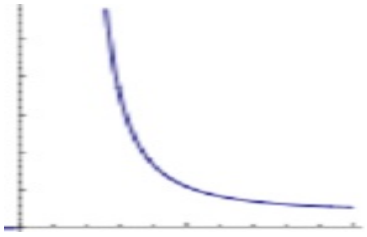


Figure 11: $\langle v_i^2 \rangle$ vs t, just to illustrate that the graph is smooth.

These suggest that we should study turbulence statistically. Therefore some statistical quantities and the notion of homogeneity and isotropy are presented in the next section.

6.3 Statistics of turbulence

6.3.1 Some properties turbulence

Some properties of a turbulent fluid are:

- The velocity field in a turbulent flow is random in space and time.
- Apart from the mean flow, turbulence looks the same irrespective of the geometry that produced it.
- Moreover, in grid turbulence for example, the state of the turbulent fluid looks almost the same at all places and in all directions at sufficiently high $\mathfrak{R}$, as the wake size increases with $\mathfrak{R}$. Some anisotropy is due to the mean flow in the fluid.
- It is also known that turbulent fluid consists of eddies of various sizes.

Let the largest eddies be of the size L. This scale is called the *integral scale*. The fluid then consists of many eddies of size L within which are present smaller and smaller eddies. The turbulence is *homogeneous* as eddies of any size are randomly or uniformly distributed. Any volume of length scale larger than L is sure to capture all the information in the turbulent fluid, since it is sure to contain eddies of all sizes. Now, the notion of mean, homogeneity and isotropy has to be formulated statistically.

Let a velocity component at a particular place be written as:

$$v_i = \langle v_i \rangle + u_i$$

Hence,

$$\langle u_i \rangle = 0$$

The average kinetic energy per unit mass is

$$\begin{aligned} \frac{1}{2}\langle \vec{v}^2 \rangle =& \frac{1}{2}\langle (\langle \vec{v} \rangle^2 + 2\langle \vec{v} \rangle \cdot \vec{u} + \vec{u}^2) \rangle \\ =& \frac{1}{2}(\langle \vec{v} \rangle^2 + \langle \vec{u} \rangle^2) \end{aligned} \tag{9}$$

The fact that $\langle \vec{v} \rangle$ is the mean flow can be justified after we know about statistical properties like homogeneity and isotropy. For now, let $\langle \vec{v} \rangle$ be accepted as the mean flow. In (9), there is no cross term between mean flow and fluctuating flow as far as KE is concerned. Thus, we can only consider the fluctuating part leaving out the mean flow, which removes at least some anisotropy of the geometry in the flow. Any velocity referred to from now on will be the turbulent velocity, $\vec{u}$.

6.3.2 Correlation function

Consider the integral $\int_{-\infty}^{\infty} f(x)g(x+r)dx$. Roughly, this gives a large positive value if the trend of f matches with that of g shifted by r i.e. if the peaks of f and g occur together, if f and g are on the same side of the y axis simultaneously for most x.

Now consider $\langle f(x)g(x+r)\rangle$. This is just a discrete version of the above type. If f and g are *random* (can take any positive and negative values) and do not reflect processes influencing each other, $f(x)g(x+r)$ in any realization has equal chances of being positive or negative. Hence, over many realizations,

$$\langle f(x)g(x+r)\rangle = 0$$

$\langle f(x)g(x+r)\rangle$ therefore gives a measure of how f and g are related to each other for a particular relative shift r.

$\langle f(x)g(x+r)\rangle$ is called the correlation function (between f and g at respective space time coordinates)

Consider $\langle u_i^2(\vec{x},t)\rangle$ Now, if the turbulence is homogeneous, there is no reason why it should depend on $\vec{x}$. If it is also isotropic, we may expect that it is independent of the orientation of the coordinate system, hence for any $i = x, y$ or z.So, let

$$\langle u_i^2(\vec{x},t)\rangle = u^2$$

The velocity correlation function Q_{ij} is defined as

$$Q_{ij} = \langle u_i(\vec{x},t)u_j(\vec{x}+\vec{r},t)\rangle$$

It is in general a function of $\vec{x},\vec{r}$ and t.

Again, if the turbulence is homogeneous, there is no reason why it should depend on $\vec{x}$. Furthermore if it is also isotropic, we may as well expect that it is independent of the direction of $\vec{r}$, since for a given $\vec{r}$, we can always orient, say ,our x axis along $\vec{r}$. If the whole process is stationary in time, then we may assume that this statistical property is independent of time. Hence, we may be allowed to write

$$\begin{aligned} Q_{xx} &= \langle u_x(\vec{x})u_x(\vec{x}+\vec{r})\rangle = u^2 f(r) \\ Q_{xy} &= \langle u_x(\vec{x})u_y(\vec{x}+\vec{r})\rangle = u^2 g(r) \end{aligned} \tag{10}$$

If r is larger and larger, we may expect $Q_{ij} \to 0$ since they will correspond to largest eddies of different domains. The typical shape of f and g are as shown in fig(12). This allows us to approximate the integral scale as the length of the rectangle of breadth 1 and having the same area as that of f:

$$\lambda_f = L = \int_0^{\infty} f(r)dr$$

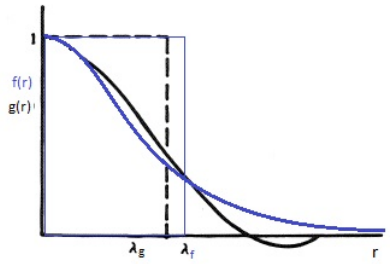


Figure 12: Typical shape of correlation functions and their means.

6.3.3 Relation between different means

In homogeneous and isotropic turbulence, the mean flow can be measures by time average, spatial average or ensemble average. The reason may be as follows.

- **Time average**

$$\overline{v_i} = \frac{1}{T}\int_t^{t+T} v_i dt$$

  is independent of t when the flow is steady on average. This gives the mean flow when T is about the order of the largest eddies (Large eddies have large turnover times). In T, the system becomes the same as before since the larger eddies would have broken down to and replaced the state of the smaller eddies in the hierarchy and new large eddies would have been created by the mean flow.

- **Spatial average**

$$\text{avg}(v_i) = \frac{1}{L^3}\int v_i d^3x$$

  Where L is the integral scale. At a particular instant of time, a volume of this size would contain all different states a fluid element would exhibit at a particular point in space over time T.

- **Ensemble average**

  Since the system is nonlinear, small variations in the initial conditions lead to drastically different outcomes. The ensemble average is the average over all these possible states of a fluid element at a particular position and time. But all possible states are contained at any instant of time in a L^3 volume of space. Hence ensemble average equals spatial average.

6.3.4 Structure function

Suppose we wanted to know roughly the cumulative energy density (per unit mass) of eddies of less than a given size. As a rough estimate, we can go to the frame of reference of any fluid element and measure the relative velocity of a neighboring

element at a distance r. This element would be the center of rotation of the neighboring elements belonging to vortices of radius r. It may also capture the relative velocity of diametrically opposite arms of eddies of size less than r. It also captures the velocity gradient of large eddies of size greater than r. Thus,

$$\begin{aligned}\langle \Delta\vec{v}^2 \rangle =& \langle [\vec{v}(\vec{x}) - \vec{v}(\vec{x}+\vec{r})]^2 \rangle \\ =& 6u^2 - 2\langle \vec{v}(\vec{x}) \cdot \vec{v}(\vec{x}+\vec{r}) \rangle \\ =& 6u^2 - 4R(r)\end{aligned}$$

Where

$$\begin{aligned}R(r) =& \frac{1}{2}\langle \vec{u}(\vec{x}) \cdot \vec{u}(\vec{x}+\vec{r}) \rangle \\ =& \frac{1}{2}\langle u_i(\vec{x}) u_i(\vec{x}+\vec{r}) \rangle \\ =& \frac{1}{2}(Q_{xx} + 2Q_{xy}) \\ =& \frac{1}{3}\vec{v}^2(\frac{f}{2} + g)\end{aligned} \tag{11}$$

As $r \to \infty$, $R(r) = 0$ and $\langle \Delta\vec{v}^2 \rangle = \frac{4}{3}KE$. Hence we expect that the structure function gives a measure of:

$\frac{4}{3}$(energy contained in eddies less than r) + r^2(sum of $\vec{\omega}^2$ of all eddies greater than r)

Now, this is a hint that the structure and velocity correlation functions are somehow related to the KE per unit mass contained in eddies of a particular size. So, it would be nice to somehow distinguish between eddies of different sizes and find their contribution to the total energy content in the turbulent fluid.

6.4 Kolmogorov Theory

6.4.1 Richardson energy cascade and the Turbulent state

The idea of Kolmogorov theory energy cascade is the following:

- Largest eddies are produced by the mean flow.
- Due to instability of the eddies, the large eddies breakup into smaller ones. This happens at every scale. This process is called *Richardson Energy cascade* (fig 13). The kinetic energy in larger eddies is distributed to smaller ones.
- Since statistical properties of turbulent fluids are stationary in time, there is something like a 'dynamic equilibrium' in the cascade process. Although energy is cascading down various scales, the rate of energy input to the larger scales is equal to the energy dissipated. Thus, there is no accumulation of energy with time at any scale. The energy transfer rate is independent of the length scale. This is a constant for this system. Let it be denoted by ε.

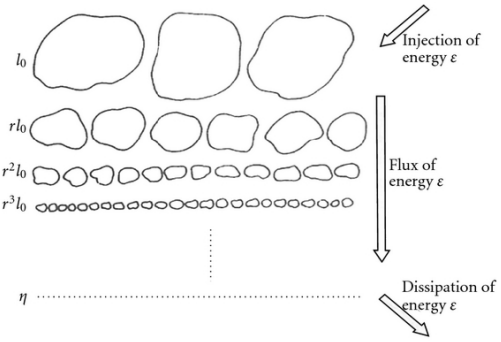


Figure 13: Richardson energy cascade.

- The energy cascade is assumed to be mostly from larger scales to smaller scales, not the other way around.
- Dissipation of energy is due to viscosity. Roughly, the eddies are not 'continuously distributed' in space. There exist regions of no eddies and the spatial distribution of the eddies is rapidly and chaotically changing with time. So the vorticity field is not a smooth one. There exist large gradients of eddies.

  The hypothesis is that, in the intermediate length scales called inertial sub range (l not too large nor too low) where the cascade is happening, ε does not depend on ν since there is very less dissipation at these scales. i.e all KE of large eddies is passed on to smaller ones in cascade.

Now, using very primitive physical arguments and dimensional analysis, we can arrive at a rough, experimentally verified universal law estimating the shape of energy distribution (with length scale) in any turbulent fluid. This is one of the most famous and significant knowledge existing in the theory of turbulence. It was proposed by A.N. Kolmogorov in 1941.

6.4.2 Scaling arguments

Assumption: Any physical quantity, say energy cascade rate, we wish to discuss at a particular scale will be of the order of the the respective length and time and velocity at that length scale. (hereafter referred to by A)

Hence, at any length scale l, we can form the following quantities:

- The characteristic velocity u of fluid in the eddies of that length scale.
- **Turnover time** t, the time it roughly takes for the eddies of this size to breakup and pass energy onto smaller ones. By using A,

$$t \sim \frac{l}{u}$$

- We can define a $\mathfrak{R}_l$ for every length scale as

$$\mathfrak{R}_l = \frac{ul}{\nu}$$

- The KE (per unit mass) of eddies at this scale $\sim u^2$
- **The energy dissipation rate**
  Since eddies breakup in time $t \sim \frac{l}{u}$, the KE in this eddy is transferred to the next scale. Hence,

$$\varepsilon \sim \frac{u^2}{t} = \frac{u^3}{l}$$

  The energy dissipation rate per unit mass from (8) is $2\nu\sigma_{ij}\sigma_{ij}$. Since σ_{ij} contains gradient of velocity, it may be correct to assume it to be of the order of $\frac{u_d}{l_d}$, since at the dissipation scale (denoted by subscript d), the velocity and vorticity are almost discontinuous due to rapid variations (see bubble analogy). Hence, Dissipation rate is also

$$\varepsilon \sim \nu\sigma_{ij}\sigma_{ij} \sim \nu\frac{u_d^2}{l_d^2}$$

  Equating the two,

$$\begin{aligned}\varepsilon \sim \nu\frac{u_d^2}{l_d^2} &\sim \frac{u^3}{l}\\ \Rightarrow \frac{\nu}{l} &\sim u\\ \Rightarrow \frac{lu}{\nu} &\sim 1\\ \text{OR}&\\ \mathfrak{R}_d &\sim 1\end{aligned}$$

  i.e Dissipation happens at very low $\mathfrak{R}$.

Since ε is a constant, we immediately have the following results:

- $$u \sim \varepsilon^{\frac{1}{3}} l^{\frac{1}{3}}$$

- **Ratio of integral scale to dissipation scale** is

$$\frac{L}{l_d} \sim \mathfrak{R}^{\frac{3}{4}}$$

- **KE is distributed as**

$$KE \sim u^2 \sim \varepsilon^{\frac{2}{3}} l^{\frac{2}{3}} \tag{12}$$

This is the famous **two thirds law of Kolmogorov**.

It is not possible to be any more quantitative than this. However, we can establish some relations between the Energy spectrum, which is the kinetic energy per unit mass per unit $\delta\frac{1}{l}$ contained in eddies whose scale lies between l and $l + \delta l$, and the structure functions. This will involve using Fourier transformation as an attempt to filter the velocities according to eddy sizes.

6.5 Kolmogorov Spectrum in Fourier space

As far as physics of Kolmogorov theory is concerned, it is the content of the previous section. However, we can be little more precise theoretically.

6.5.1 The Fourier transform as a filter

Consider uniformly distributed eddies in space, in a planar fashion (like rotating cylinders of fluid of diameter l, with its axis as z axis). v_x for eg. varies along a diameter of a cylinder as follows:

- first it is positive maximum.
- then goes to 0.
- then is a minimum (negative) with same modulus as maximum.

This, due to uniform spacing of eddies, approximately behaves like a sine function. A Fourier transform of this velocity field would have a steep peak at $k = \frac{2\pi}{l}$. Thus, although the velocity along the diameter varies as $\frac{1}{r}$, the Fourier transform is a good filter to differentiate between different eddies.

What the Fourier transform of the velocity field does, is to consider the wave number as $k = \frac{2\pi}{l}$ where l is the diameter of the eddies. It assumes that eddies of a particular wavenumber are uniformly distributed in space, although different modes can be placed out of phase with each other. This would allow us to define properly the distribution of KE across wavenumbers.

6.5.2 One dimensional spectra

Although we know roughly the relation between the structure function and Energy function, it would be nice to know the interrelation between the correlation function, the structure function and the Energy function (in real space or Fourier space. Only the Fourier space version of the energy function has been learnt in this project).

Consider a one dimensional equivalent of (10). Let $g(x)$ have a form of

$$g(x) = A_1 sin(k_1 x + \phi_1) + A_2 sin(k_2 x + \phi_2) \tag{13}$$

i.e, the velocity is the superposition of sinusoidal velocities with velocity amplitudes A_1 and A_2 respectively. The KE is:

$$g^2(x) = A_1^2 sin^2(k_1 x + \phi_1) + A_2^2 sin^2(k_2 x + \phi_2) + 2A_1 A_2 sin(k_1 x + \phi_1) sin(k_2 x + \phi_2)$$

Thus the energy contribution is A_1^2 from mode k_1, A_2^2 from mode k_2 and a cross term contribution $A_1 A_2$, which contributes nothing when we consider a spatial average over a suitable interval (since it is a cross term. It has an average of 0 while the others have an average of half).

Correlation is theoretically achievable by spatial integration (over length scale larger than the largest eddies and smaller than the size of the system) instead of ensemble averaging due to the supposed equivalence of different methods of averaging.

Also, due to homogeneity, the autocorrelation (in 1 dim) is defined in terms of the convergent integral given below:

$$h(r) = \frac{1}{2a}\int_{-a}^{a} u_x(x)u_x(x+r)dx$$

The autocorrelation (cross correlation of a function with itself) coefficient is defined as:

$$f(r) = \frac{\int_{-a}^{a} u_x(x)u_x(x+r)dx}{\int_{-a}^{a} u_x^2(x)dx} \tag{14}$$

In our case (13), we may let $a \to \infty$ as g does not die out at large a. This comes out to be

$$h(r) = A_1^2 cos(k_1 r) + A_2^2 cos(k_2 r)$$

The Fourier transform of this will have delta functions of coefficients A_i^2. Thus, the Fourier transform of the correlation coefficient gives the energy contribution (per unit k width) of a particular mode.This is called the *One dimensional spectrum* of a random function. More generally, if $G(k)$ is the FT of $g(x)$, then,

$$FT\int_{-\infty}^{\infty} g(x)g(x+r)dx = 2\pi G(k)G^*(k) \tag{15}$$

Provided g is square integrable. This is the *One dimensional spectrum* of $g(x)$.

If u_x is not square integrable, it is first clipped:

$$g_a(x) = \begin{cases} g(x) & \text{if } |x| < a, \\ 0 & \text{if } |x| > a. \end{cases}$$

This has (15) well defined. The equation is then ensemble averaged, divided by $2a$ and a is let $\to \infty$

$$FT.\langle g_a(x)g_a(x+r)\rangle = \lim_{a\to\infty}\frac{\pi}{a}\langle G_a(k)G_a^*(k)\rangle$$

LHS is independent of x and the relation reduces to:

$$FT.\langle g(x)g(x+r)\rangle = \frac{\pi}{a}\langle G_a(k)G_a^*(k)\rangle \tag{16}$$

Although $a \to \infty$, RHS remains finite (see [10] for details).

6.5.3 Spectrum in 3 dimensions

The autocorrelation function in 3 dimensional is R(r) defined in (11). It is achievable theoretically, analogously to (14) in the homogeneous case, in the following way

$$\begin{aligned}\langle \vec{u}(\vec{x}) \cdot \vec{u}(\vec{x}+\vec{r}) \rangle =& \frac{1}{V} \int_V \vec{u}(\vec{x}) \cdot \vec{u}(\vec{x}+\vec{r}) d^3x \\ =& R(r)\end{aligned}$$

Length scale of volume V is chosen appropriately so that it agrees with the ensemble average.

Analogous to the previous section, we can clip functions and integrate over cubes of side 2a.

The three dimensional equivalent of equation (16) is:

$$FT.[R(r)] = \lim_{a\to\infty} \frac{\pi^3}{a^3} (G(k)G^*(k))$$

Where FT. here is the three dimensional Fourier Transform.

$$\begin{aligned}\frac{1}{(2\pi)^3} \int [R(r)] e^{i\vec{k}\cdot\vec{x}} d^3x = & \lim_{a\to\infty} \frac{\pi^3}{a^3} (G(k)G^*(k)) \\ =& A(\vec{k})\end{aligned} \tag{17}$$

But $A(\vec{K})$ is not what is called the 3 dimensional spectrum $E(k)$. Because of homogeneity and isotropy, $A(\vec{k})$ is only $A(k)$ where $k = |\vec{k}|$. $E(k)$ is defined through the following relation:

$$\begin{aligned}\iiint A(k) d^3k =& \int (A(k) 4\pi k^2) dk \\ =& \int E(k) dk\end{aligned}$$

$A(k)$ is simply integrated over θ and ϕ after going to spherical coordinates in Fourier space

The RHS of Fourier transform (17) is also simplified in a similar manner (in real space) and in terms of E(k) becomes:

$$E(k) = \frac{2}{\pi} \int_0^\infty R(r) sin(kr) kr dr$$

The inverse transformation is given by:

$$R(r) = \int_0^\infty E(k) \frac{sin(kr)}{kr} dk$$

It can be seen that when $r \to 0$,

$$\frac{1}{2}\langle \vec{u}^2 \rangle = \int_0^\infty E(k)dk$$

Hence, $E(k)$ is interpreted as giving the *Energy contribution from eddies whose size belong to the interval* $(k, k+\Delta k)$*, per unit interval* Δk.

6.5.4 Five thirds law

Equation (12) can be now written in terms of k as:

$$KE \sim \varepsilon^{\frac{2}{3}} k^{-\frac{2}{3}}$$

By *dimensional analysis* we see that $E(k)k$ has to go as KE. Hence,

$$E(k) \sim \varepsilon^{\frac{2}{3}} k^{-\frac{5}{3}}$$

This is the famous **five thirds law** of turbulence.
This has been experimentally verified to hold in the inertial subrange.

$$l_d < l < L$$

An experimental plot of Kolmogorov spectrum is shown in fig(14) below.

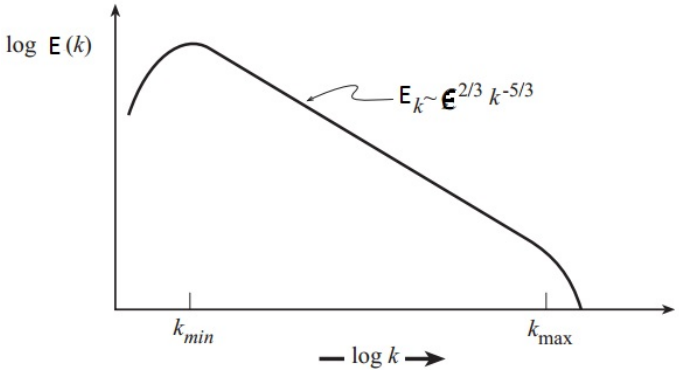


Figure 14: Schematic of experimental plot of E(k).

7 Diffusion

Now that turbulence has been learnt semi quantitatively, the phenomenon of turbulent diffusion can now be attempted to study.

7.1 Molecular diffusion

Molecular diffusion is the thermal motion of all (liquid or gas) particles at temperatures above absolute zero. The rate of this movement is a function of temperature, viscosity of the fluid and the size (mass) of the particles. Diffusion explains the net flux of molecules from a region of higher concentration to one of lower concentration.

An example is perfume in air. The scalar (molecules of perfume) are present in a flowing fluid in general (air). To a good approximation, we can consider the perfume molecules as a passive additive i.e. its presence does not affect the flow of the fluid. The air in a closed room is not flowing, so the concentration of the scalar is governed by molecular diffusion. In general, the scalar additive can be diffusing through a flowing fluid, and the molecular diffusion governs the concentration distribution as long as the flow is laminar. This is reasonable, since in a suitably chosen frame of reference, most of the fluid will be almost at rest.

The additive can be considered as another passive fluid, with concentration $C(\vec{x},t)$, not interacting with the fluid medium. The dynamics of C is governed by the dynamics of the fluid medium. Thus, the additive is considered to be frozen inside the fluid medium.

Let us first assume that the fluid medium is at rest. To obtain the dynamics of the additive fluid we impose the fact that the fluid flows in such a manner that its mass is conserved. Since mass density is proportional to concentration density, we obtain

$$\nabla \cdot \vec{J} + \frac{\partial C}{\partial t} = 0 \tag{18}$$

where

$$\vec{J} = C\vec{v}_a$$

The velocity $\vec{v}$ is the velocity of the additive fluid.

Molecular diffusion is governed by Fick's law, which postulates that the flux goes from regions of high concentration to regions of low concentration, with a magnitude that is proportional to the concentration gradient. i.e,

$$\vec{J} = -k\nabla C$$

Substituting in (18) we get the Diffusion equation:

$$\frac{\partial C}{\partial t} = k\nabla^2 C$$

This equation can be solved easily by Fourier transform. Given an initial distribution $C(\vec{x},0)$, the distribution at any later time t is given by

$$C(\vec{x},t) = \frac{1}{4\pi k_m t}\int exp(-\frac{|\vec{x}-\vec{x}'|}{4kt})C(\vec{x}',0)d^3x'$$

If the initial distribution is concentrated at a point, it will spread over a length scale L in time $\tau \sim \frac{L^2}{k}$. For $k \sim 0.1\ cm^2s^-1$ we may expect perfume to diffuse across a room of size 5m in time $\tau\tau \sim 600$h. This conclusion contradicts our experience. One resolution of this problem is that the air in the room is not static. If the room is not closed i.e a the door or a window is open and there is a breeze, perfume can be *advected* along with the flow of the medium. Another idea is to see What happens if the room is sealed, but air flows randomly within. To address this problem we need to allow the fluid medium to flow randomly.

Let the velocity of the additive element be written as $\vec{v} = \vec{u}_m + \vec{u}_a$ where $\vec{u}_m$ is the velocity of the fluid medium and $\vec{u}_a$ is the velocity of the additive in the rest frame of the fluid element at $\vec{x}$. Then,

$$\vec{J} = C(\vec{u}_m + \vec{u}_a)$$

From Fick's law,

$$\vec{u}_a = -k\nabla C$$

Thus the diffusion equation becomes,

$$\begin{aligned} \frac{\partial C}{\partial t} + \vec{u}_m \cdot \nabla C &= k\nabla^2 C \\ \Rightarrow \frac{dC}{dt} &= k\nabla^2 C \end{aligned} \tag{19}$$

When the fluid is incompressible. This equation just molecular diffusion in the frame of reference of the fluid element.It is called the *Advection Diffusion equation*.

Now, we need to see what happens when the velocity field of the medium is chaotic and turbulent.

7.2 Turbulent diffusion of a passive scalar

Following the statistical method followed in turbulence, what is required are the evolution of averaged quantities since only they are reproducible in an experiment. The velocity field of the medium is to be written as

$$\vec{v} = \langle\vec{v}\rangle + \vec{u}$$

From now on, the subscript m in $\vec{u}_m$ will be dropped and the mean velocity of flow will be assumed to be 0. Since we are interested in turbulent diffusion in

a closed room. Also, the turbulent velocity (which is the total velocity) will be denoted by $\vec{u}$ as before.

The incompressibility of flow implies:

$$\nabla \cdot \vec{v} = \nabla \cdot \vec{u} = 0$$

Also

$$\langle \vec{u} \rangle = 0$$

The concentration is similarly decomposed into a sum of mean and turbulent concentrations:

$$\begin{aligned} C =& \langle C \rangle + c \\ =& \overline{C} + c \end{aligned}$$

The ensemble-averaged advection diffusion equation (19) then becomes

$$\frac{\partial \overline{C}}{\partial t} + \nabla \cdot \langle c\vec{u} \rangle = k\nabla^2 C \tag{20}$$

Now we need to estimate $\langle c\vec{u} \rangle$ in order to close this equation, since $\langle c\vec{u} \rangle$ is a new variable we produced due to ensemble averaging. Generally, we must solve the corresponding equation of the fluctuating concentration c and substitute back. To obtain the dynamical equation for C, we need to subtract this from equation (19):

$$\frac{\partial c}{\partial t} + \vec{u} \cdot \nabla \overline{C} + \nabla \cdot (c\vec{u} - \langle c\vec{u} \rangle) = k\nabla^2 c$$

Major contribution to this equation might be from the second term since $(c\vec{u} - \langle c\vec{u} \rangle)$ may be of order smaller than c. Also, its ensemble average is zero i.e.

$$\begin{aligned} \langle (c\vec{u} - \langle c\vec{u} \rangle) \rangle =& (\langle c\vec{u} \rangle - \langle c\vec{u} \rangle) \\ =& 0 \end{aligned}$$

, hence it is not a reproducible quantity. On the basis of this and simplicity, we may leave it.

If the turbulence in the fluid enhances the rate of diffusion, we must obtain, at least approximately, for $\langle c\vec{u} \rangle$ something like Fick's law i.e.

$$\langle c\vec{u} \rangle \sim -k_t \nabla \overline{C}$$

Where k_t is the turbulent diffusivity.

So, here successive approximations are to be made, to proceed in this direction:

- The molecular diffusion coefficient is very small when compared to the advection term, as was seen when the approximate time for perfume to diffuse was calculated. Hence, we may drop the molecular diffusion term on the RHS.

$$\frac{\partial c}{\partial t} = -\vec{u} \cdot \nabla \overline{C}$$

This is easily solved by:

$$c(\vec{x},t) = -\int_0^t u_j(\vec{x},t') \frac{\partial}{\partial x_j} \left(\overline{C}(\vec{x},t')\right) dt'$$

Therefore,

$$\begin{aligned} cu_i &= -u_i(t) \int_0^t u_j(t') \frac{\partial}{\partial x_j} \left(\overline{C}(t')\right) dt' \\ &= -\int_0^t u_i(t) u_j(t') \frac{\partial}{\partial x_j} \left(\overline{C}(t')\right) dt' \end{aligned}$$

The $\vec{x}$ dependence has been suppressed. Taking ensemble average on both sides,

$$\begin{aligned} \langle cu_i \rangle &= -\int_0^t \langle u_i(t) u_j(t') \frac{\partial}{\partial x_j} \left(\overline{C}(t')\right) \rangle dt' \\ &= -\int_0^t \langle u_i(t) u_j(t') \rangle \frac{\partial}{\partial x_j} \left(\overline{C}(t')\right) dt' \end{aligned} \tag{21}$$

Since $\overline{C}$ is already an ensemble averaged quantity.

- When turbulence is statistically stationary in time, the correlation function $\langle u_i(t) u_j(t') \rangle$ depends on only $|t' - t|$.

Analogous to the two point velocity correlation for functions separated in space, this is the two point correlation in time. Thus, it is written as:

$$\langle u_i(t) u_j(t') \rangle = u^2 G_{ij} \tag{22}$$

where

$$G_{ij} = G_{ij}(t' - t)$$

G_{ij} is the correlation coefficient, which tells something about how velocities at different times are correlated.

The integral (21) becomes:

$$\langle cu_i \rangle = -\int_0^t u^2 G_{ij} \frac{\partial}{\partial x_j} \left(\overline{C}(t')\right) \rangle dt' \tag{23}$$

- In the case of isotropic turbulence, we may expect the symmetric tensor G to be isotropic.

i.e of the form

$$G_{ij} = g(t' - t)\delta_{ij} \tag{24}$$

Thus, the correlation involving different velocity components is assumed to be zero in this case. Let us assume it to be so. (this correlation function in time g has nothing to do with the previous correlation function Q_{xy}). Its shape (fig.14 below) is similar to f in figure **??**, with $t' - t$ on the x axis instead of r.

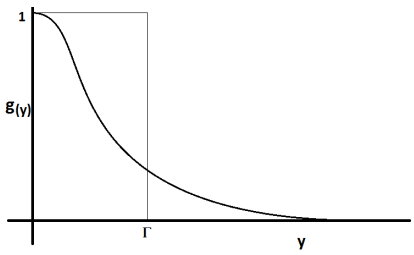


Figure 15: The autocorrelation coefficient.

Let its mean width called *correlation time* be defined similarly as before:

$$\tau = \int_0^\infty g(y)dy$$

With

$$y = (t' - t)$$

- It is observed that τ is usually very small. Since g falls rapidly with y, when $|y| > \tau$, it would have a considerably low value and hence it would be a good approximation to clip the function after $|y| > \tau$. This reduces the limits on the integral (23) to:

$$\langle cu_i \rangle = -\int_{t-\tau}^{t} u^2 g(t' - t) \frac{\partial}{\partial x_i} \left(\overline{C}(t') \right) dt'$$

- Further, due to *Reynold's rules of averaging*,

$$\overline{\frac{\partial C}{\partial x_i}} = \frac{\partial \overline{C}}{\partial x_i}$$

We know that ensemble averages can also be interpreted as time averages. Hence, we can make an approximation that $\frac{\partial \overline{C}}{\partial x_i}$ does not vary very much during the interval τ. We can then replace it in the integral by its constant value evaluated at t:

$$\langle cu_i \rangle = -\int_{t-\tau}^{t} u^2 g(t'-t)dt' \quad \frac{\partial \overline{C}}{\partial x_i}(t)$$

This is now in the form of Fick's. i.e

$$\langle c\vec{u} \rangle = -\int_{t-\tau}^{t} u^2 g(t'-t)dt' \quad \nabla \overline{C}(t)$$

The complete equation (20) after all these approximations becomes:

$$\frac{\partial \overline{C}}{\partial t} - k_t \nabla \cdot \nabla \overline{C} = k\nabla^2 C$$
$$\Rightarrow \frac{\partial \overline{C}}{\partial t} = (k_t + k)\nabla^2 C$$

Where we can define

$$k_t = \int_{t-\tau}^{t} u^2 g(t'-t)dt' \tag{25}$$

as *turbulent diffusivity*. Thus, turbulence in fluids helps diffuse the additives.

To see by how much it increases the rate of diffusion, let us estimate (25). If g rapidly falls, we can assume it to have a value of 1 throughout the time interval τ. Hence,

$$k_t \sim u^2 \tau$$
$$\sim \frac{1}{3} v^2 \tau$$

Where v is the total RMS velocity i.e. $v = \langle \vec{u}^2 \rangle$.

For typical velocity fluctuations, $v \sim 10\ cm\ s^{-1}$ with correlation time $\tau \sim 30s$, we have $k_t \sim 1000\ cm^2\ s^{-1}$, $k_t \sim 10^4 k$. Hence, the perfume diffuses across 5m in 4 minutes, which is much more reasonable than our previous estimate which was based on just molecular diffusion.

Hence, Turbulence indeed does enhance diffusion. But in some practical cases, turbulence is anisotropic and inhomogeneous. The correlation tensor (22) then has to be treated as a symmetric tensor and cannot be simplified as (24). Similar approximations can be followed starting from (23) to get an anisotropic version of the diffusion equation.

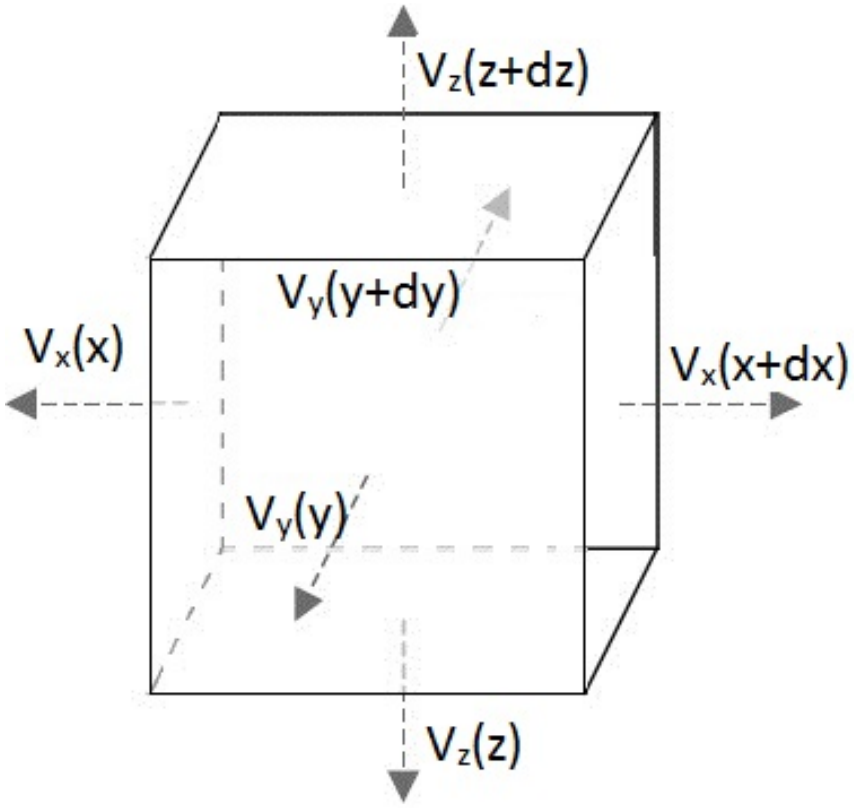
8 Summary

In this report, first the basic equations of fluid mechanics are derived.The famous Bernoulli's theorem is derived and also obtained from the Euler Equation. Viscosity is then introduced, leading to the Navier-Stokes equation. Dissipation of energy due to viscosity is derived. Then, some flows are described which take us to turbulence. Kolmogorov theory of homogeneous isotropic turbulence is discussed. With this knowledge, Turbulent Diffusion is discussed in the homogeneous isotropic case.

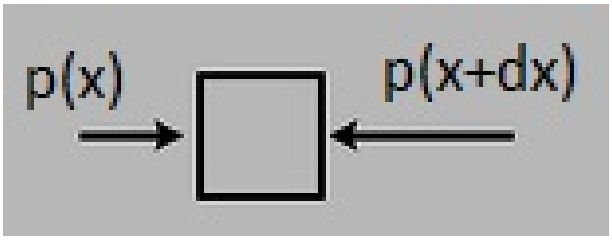
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In future, simulations can be run to verify the estimate of turbulent diffusivity.

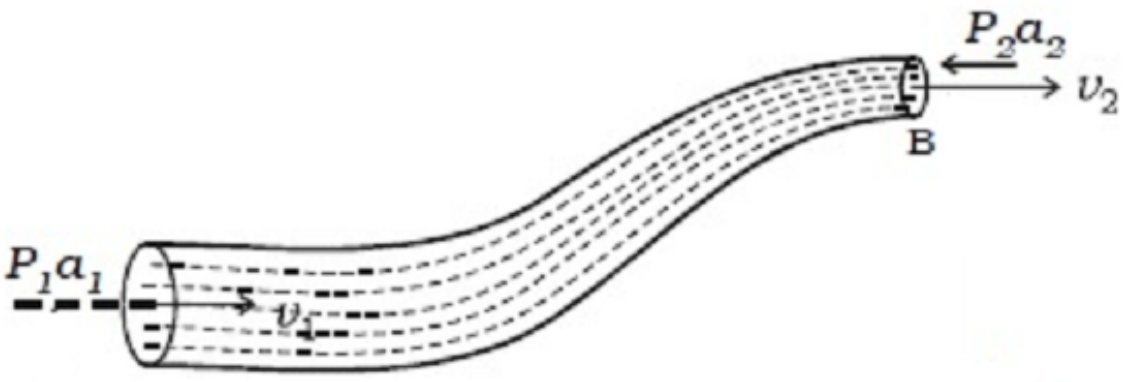
9 Figures



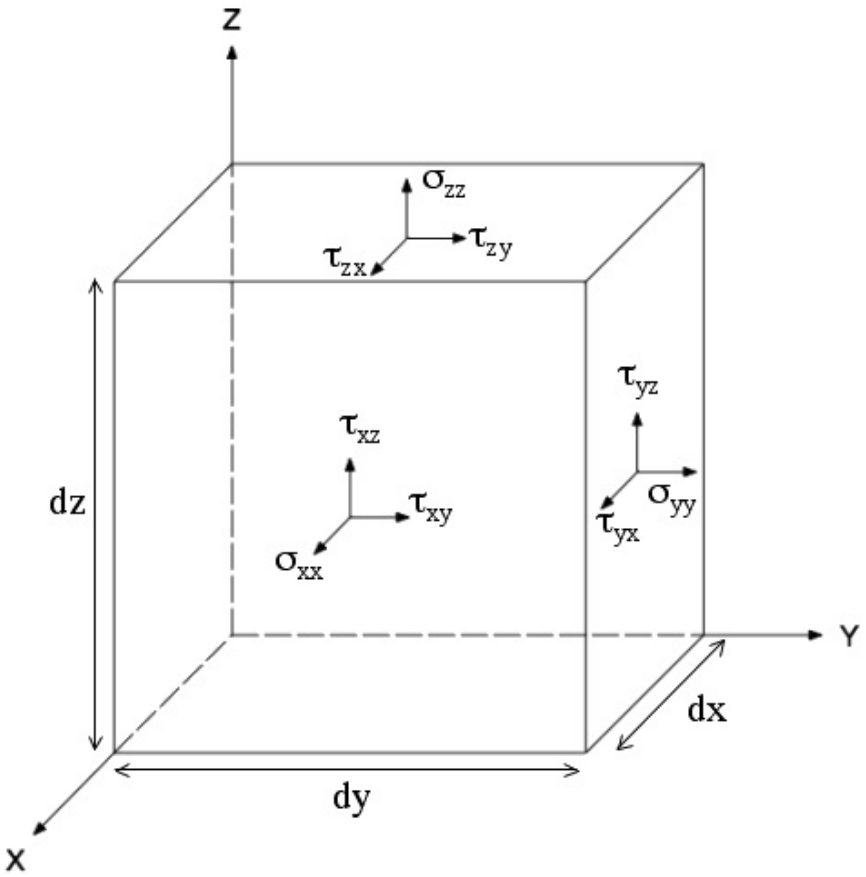
1. An expanding cube



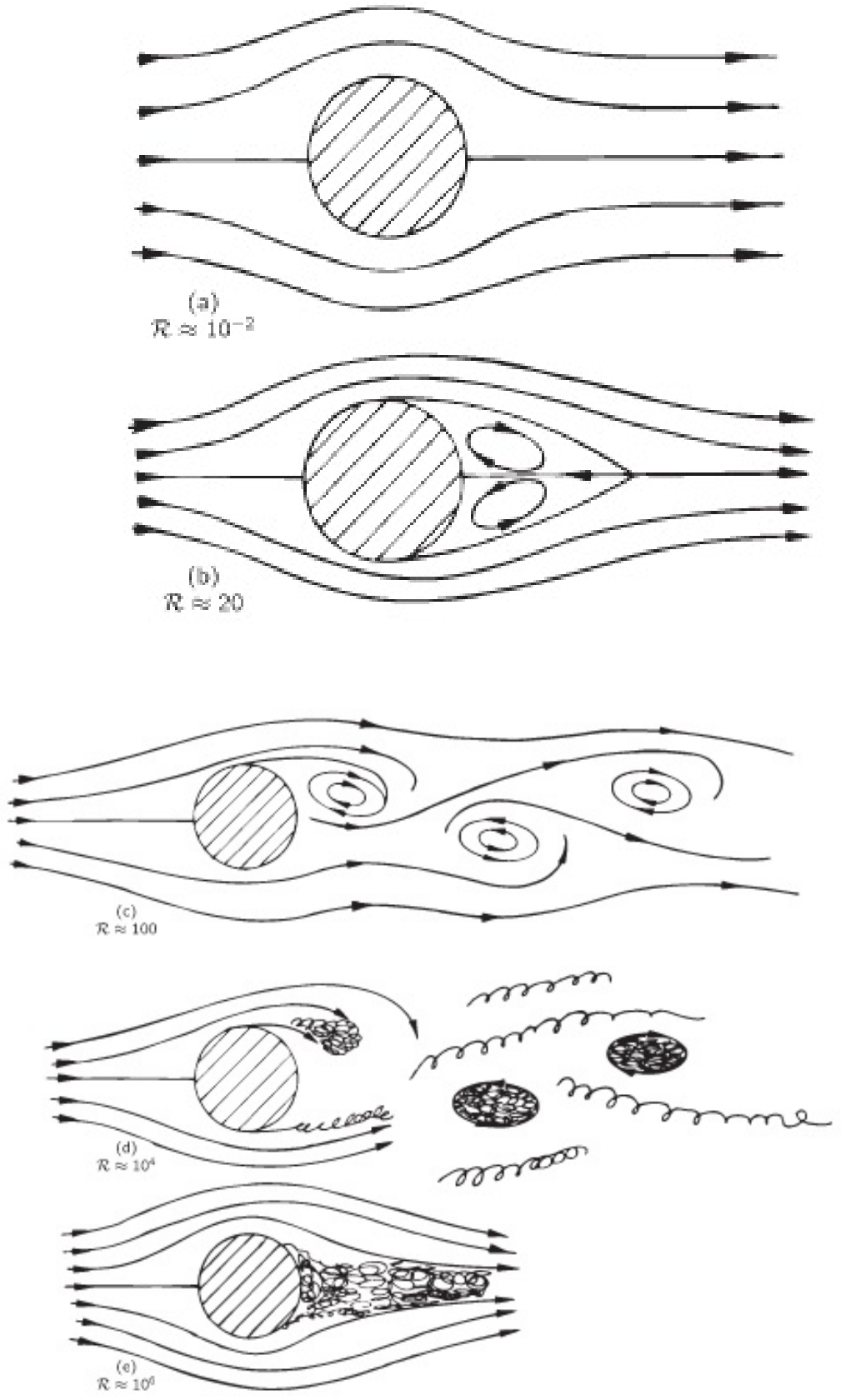
2. Force due to pressure on a cube



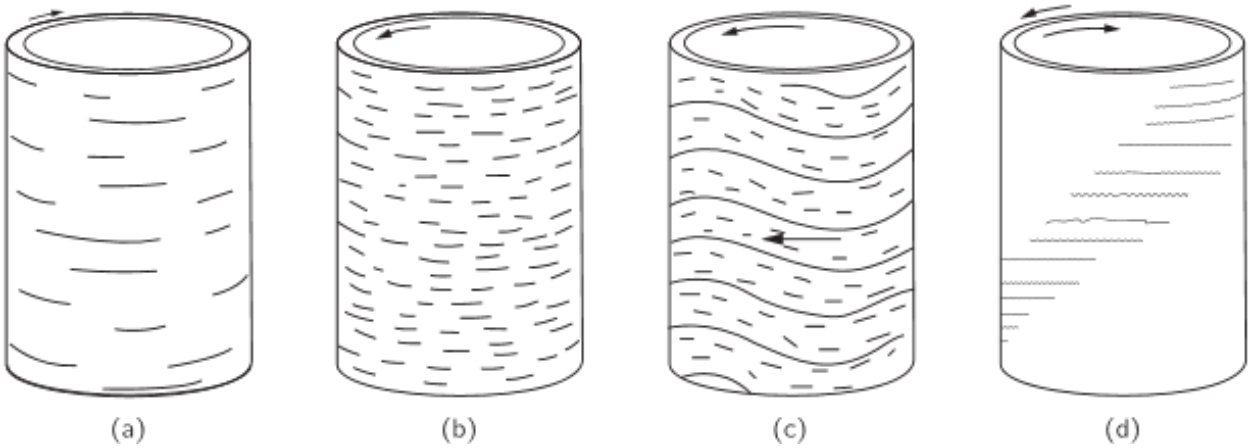
3. A stream tube



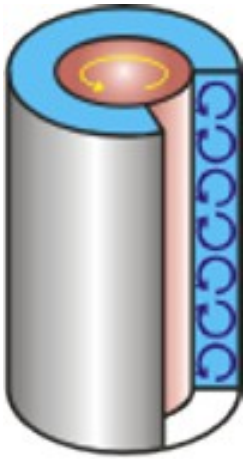
4. Viscous forces on a cube



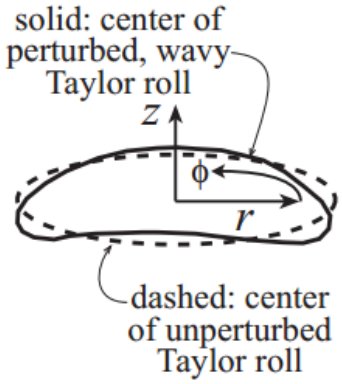
5. Transition to turbulence as $\Re$ is increased. Taken from [3]



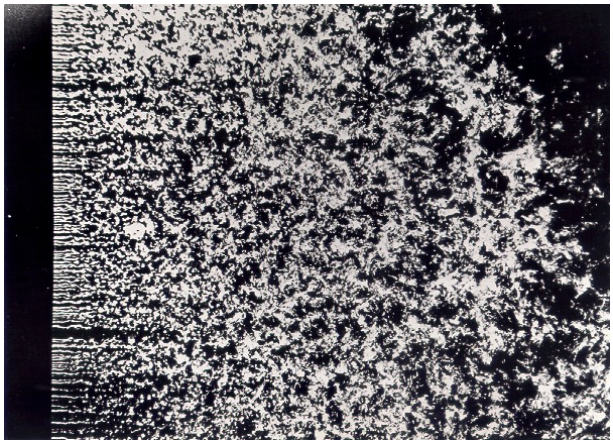
6. Couette flow, taken from [3]



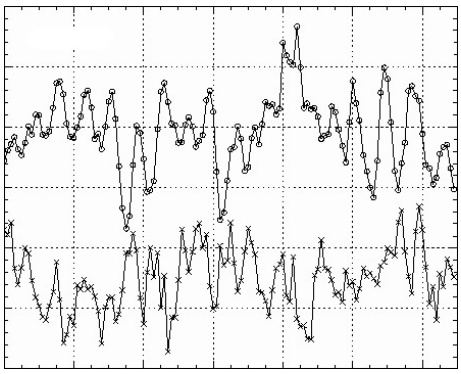
7. Taylor cells, the second phase of couette flow.



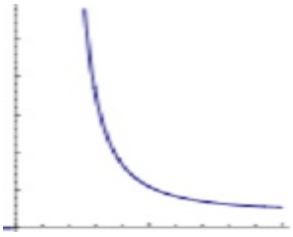
8. Wavy Taylor cells, the third phase of couette flow. Taken from [5]



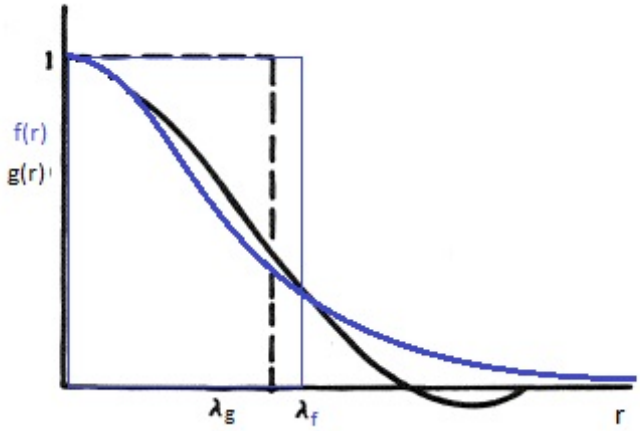
9. Turbulence behind a grid.



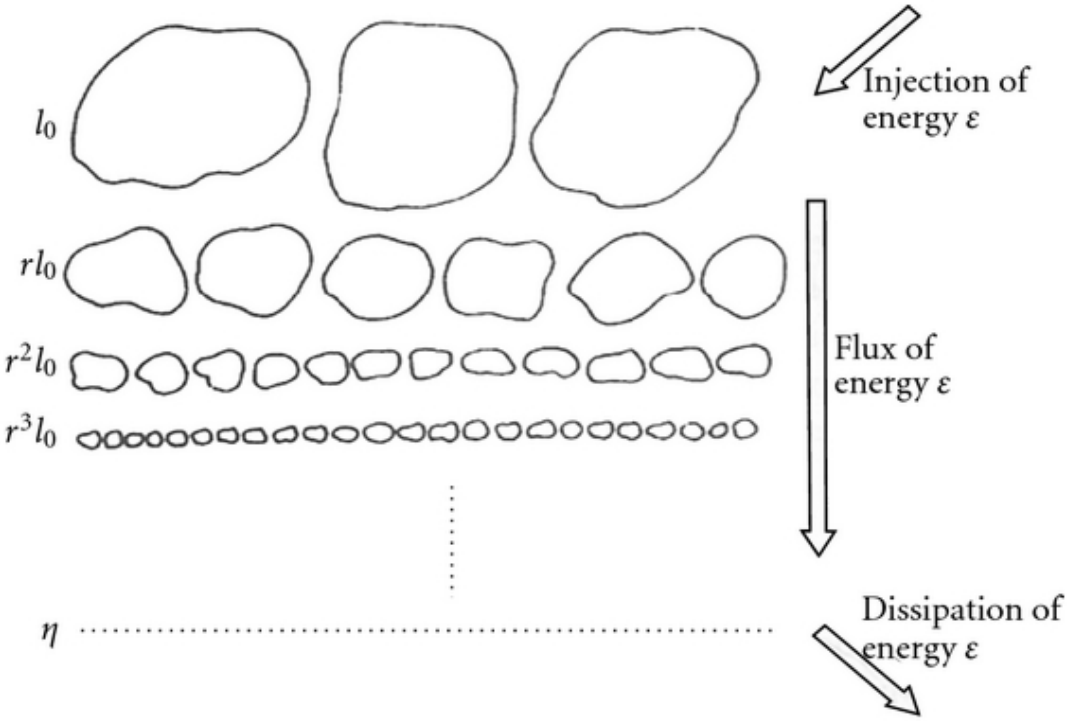
10. $v_i(\vec{x}_0, t)$ vs t for two different realizations.



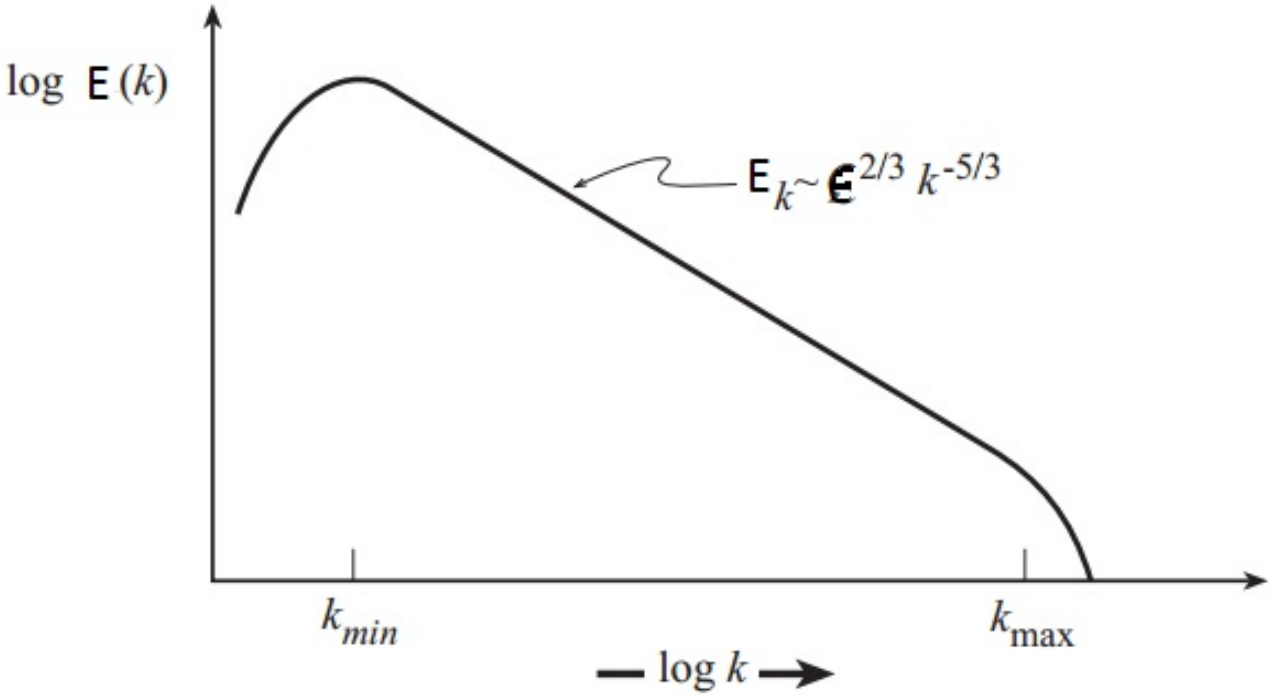
11. $\langle v_i^2 \rangle$ vs t, just to illustrate that the graph is smooth.



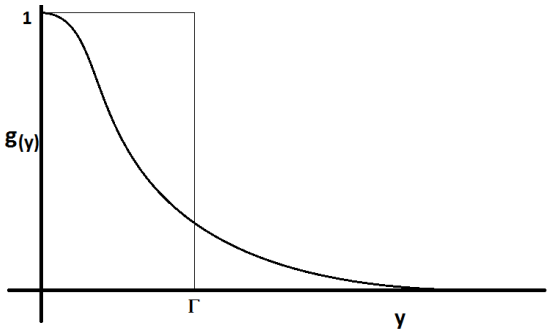
12. Typical shape of correlation functions and their means.



13. Richardson energy cascade.



14. Schematic of experimental plot of E(k).



15. The autocorrelation coefficient.

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